

DecodeLabs Data Analytics Internship

Project 1: Data Cleaning & Preparation

1. Objective

The objective of this project is to clean and prepare a raw customer sales dataset by identifying and handling missing values, duplicate records, incorrect date formats, inconsistent text values, and incorrectly formatted numerical data.

2. Project Requirements

The project focuses on data integrity and preparation. The main requirements are to identify missing or null values, remove duplicates, and correct data formats for dates, numbers, and text.

3. Dataset

The project uses a customer sales dataset containing 103 raw records and 9 columns.

Column	Description
Customer_ID	Unique customer identifier
Name	Customer name
Email	Customer email
City	Customer's city
Age	Customer age
Purchase_Date	Date of purchase
Product	Purchased product
Quantity	Quantity purchased
Purchase_Amount	Purchase amount

4. Data Cleaning Process

Missing Values

Missing values were identified using Pandas. Missing Age and Quantity values were filled using the median. Missing City values were filled with 'Unknown', and missing Email values were filled with 'Not Provided'.

Duplicate Customer IDs

Duplicate Customer_ID values were identified and duplicate records were removed while keeping the first occurrence.

Date Cleaning

Purchase_Date values were converted using Pandas date conversion with mixed-format handling. Invalid dates were identified and records with invalid dates were removed. Valid dates were standardized to YYYY-MM-DD.

Text Cleaning

City and Product values were cleaned by removing unnecessary leading/trailing spaces and standardizing capitalization.

Numeric Data Cleaning

Purchase_Amount values containing currency symbols, commas, spaces, or numeric text were cleaned and converted into numeric values.

5. Cleaning Summary

Data Quality Issue	Cleaning Action
Missing Email	Filled with Not Provided
Missing Age	Filled using median
Missing City	Filled with Unknown
Missing Quantity	Filled using median
Duplicate Customer IDs	Removed duplicate records
Mixed date formats	Converted to a consistent date representation
Invalid dates	Removed records with invalid dates
Inconsistent City text	Trimmed spaces and standardized capitalization
Inconsistent Product text	Trimmed spaces and standardized capitalization
Purchase Amount formatting	Removed currency symbols/commas and converted to numeric

6. Final Validation

The cleaned dataset was validated using Pandas to confirm that the major data-quality issues had been resolved.

Validation Check	Final Result
Missing values	0
Duplicate Customer IDs	0
Invalid dates	0

7. Tools Used

- Python
- Pandas
- Visual Studio Code
- CSV

8. Project Files

File	Purpose
DecodeLabs Project1 raw dataset.csv	Original raw dataset
cleaned_dataset.csv	Final cleaned dataset
data_cleaning.py	Python script used for cleaning and validation
README.md	Project documentation
DecodeLabs Project 1 Report.docx	Project report

9. Conclusion

The raw customer sales dataset was successfully cleaned and prepared for further analysis. Missing values, duplicate Customer IDs, invalid dates, inconsistent text formatting, and numerical formatting issues were handled. Final validation confirmed zero missing values, zero duplicate Customer IDs, and zero invalid dates.

10. Learning Outcomes

- Loading and inspecting CSV data using Pandas
- Identifying and handling missing values
- Detecting and removing duplicate records
- Working with mixed and invalid date values
- Cleaning text data
- Converting text-based numeric values into numbers
- Performing final data-quality validation
- Exporting a cleaned CSV dataset